

LAB-2 Exercises

LAB ASSIGNMENT -2 Control Flow Testing

Submission date: Week-6 (17Aug to 21Aug)

Control Flow testing: White box testing approach. Also in test cases include the methods of black box testing.

For all the below examples perform the following tasks.

- Draw the control flow graph for the program.
 - Calculate the cyclomatic complexity of the program.
 - List all the independent paths
 - Design test cases from independent paths.
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- Write a code (any language) / Pseduocode for the given requirements.

1.	A company offers a discount on bulk purchases of an item based on the number of units purchased: • 0-9 units → No discount • 10-49 units → 5% discount • 50-99 units → 10% discount • 100+ units → 15% discount										
2.	An e-commerce website provides the following discount rules based on cart value and membership: • Cart Value < ₹1,000 → No Discount • Cart Value ₹1,000 - ₹4,999 → 5% Discount • Cart Value ₹5,000 - ₹9,999 → 10% Discount • Cart Value ₹10,000+ (and membership = 'Prime') → 20% Discount										
3	An electric power distribution company charges its domestic consumers as follows: <table><tr><th>Consumption Units</th><th>Rate of Charge</th></tr><tr><td>0-100</td><td>Rs. 0.75 per unit</td></tr><tr><td>101 –300</td><td>Rs. 75 plus Rs. 1.00 per unit excess of 100</td></tr><tr><td>301 –500</td><td>Rs. 275 plus Rs. 1.50 per unit excess of 300</td></tr><tr><td>500 and above</td><td>Rs. 575 plus Rs. 1.75 per unit excess of 500</td></tr></table>	Consumption Units	Rate of Charge	0-100	Rs. 0.75 per unit	101 –300	Rs. 75 plus Rs. 1.00 per unit excess of 100	301 –500	Rs. 275 plus Rs. 1.50 per unit excess of 300	500 and above	Rs. 575 plus Rs. 1.75 per unit excess of 500
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4	Program for finding largest from three numbers
5	Pseudocode for Checking whether inputted number is Prime or not <pre> BEGIN READ number IF number <= 1 THEN PRINT "Not a Prime Number" EXIT i = 2 isPrime = TRUE WHILE i * i <= number IF number MOD i == 0 THEN isPrime = FALSE break; END IF I = i+1 END WHILE IF isPrime THEN PRINT "Prime Number" ELSE PRINT "Not a Prime Number" END IF END </pre>
6	Pseudocode for MARKSHEET CODE <pre> BEGIN PRINT "Enter the number of subjects:" READ n DECLARE marks[n] SET total = 0 PRINT "Enter the marks for each subject:" SET i = 0 WHILE i < n DO READ marks[i] total = total + marks[i] </pre>

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	<pre>INCREMENT i by 1 END WHILE SET average = total / n SET percentage = (total / (n * 100)) * 100 // Assuming each subject is out of 100 PRINT "Total Marks:", total PRINT "Average Marks:", average PRINT "Percentage:", percentage, "%" // Determine the grade IF percentage >= 90 THEN PRINT "Grade: A+" ELSE IF percentage >= 80 THEN PRINT "Grade: A" ELSE IF percentage >= 70 THEN PRINT "Grade: B" ELSE IF percentage >= 60 THEN PRINT "Grade: C" ELSE IF percentage >= 50 THEN PRINT "Grade: D" ELSE PRINT "Grade: F (Fail)" END IF END</pre>
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